

Discount Rates and Scarcity of Capital

4

Overview

The previous chapters discussed why companies should aim for integrated value creation, which involves the balancing of capitals. When aiming for integrated value creation in investment decisions, one necessarily deals with the future. This raises the subject of present values and discount rates—the basics of corporate finance. The first section of this chapter therefore addresses discount rates and the time value of money. These concepts, and the effects of changes in discount rates, are illustrated with calculation examples.

Subsequently, the determinants of discount rates are discussed, starting with government bonds as a benchmark and then adding the premia on corporate bonds and equity. This all applies to financial capital. Next, we introduce the social discount rate for social and environmental capital. The counterparty of companies' social and environmental capital is the wider society, representing current and future generations. Leading economists argue for an equal treatment of current and future generations, which implies a low social discount rate.

Finally, we show how the financial discount rate can be expanded to an integrated discount rate that can be applied to integrated value, which also includes social and environmental value. It is shown that larger environmental and social liabilities raise the integrated discount rate. Conversely, environmental and social assets lower the integrated discount rate. Chapters 12 and 13 provide an in-depth analysis of the integrated discount rate. See Fig. 4.1 for an overview of this Chapter.

Learning Objectives

After you have studied this chapter, you should be able to:

- Interpret the time value of money
- Do basic interest calculations
- Understand and apply various types of interest rates
- Analyse the determinants of (financial) interest rates and the cost of equity
- Critically assess the determinants of discounting social and environmental capital

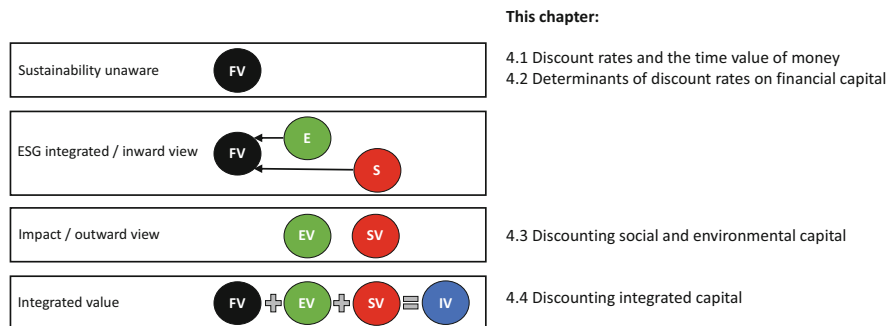


Fig. 4.1 Chapter overview

- Differentiate between discounting financial capital and discounting integrated capital

4.1 Discount Rates and the Time Value of Money

Demand and Supply of Financial Funds

Discount rates of financial capital are determined in a setting of markets with supply and demand of financial funds. Figure 4.2 illustrates the flow of financial funds from investors to users through financial markets and intermediaries (such as banks and institutional investors). A large supply of funds relative to demand lowers the price or discount rate of financial capital, *ceteris paribus*. Next, financial markets are influenced by government policies. While central banks can supply or withdraw (short-term) funds, governments make regulations to ensure a proper functioning of financial markets. Examples of such government regulations are protection of property rights, enforcement of contracts, transparency of price discovery, and supervision of financial markets and intermediaries (De Haan et al., 2020).

Time Value of Money

Most of the time, people prefer money today over money tomorrow, because of inflation and opportunity costs. This is called the *time value of money*: the difference in value between money now and money in the future. So, how to compare €100 today with €100 in 1 year? The difference is calculated by means of the *discount rate*, which is the interest rate r used to determine the *present value* PV of future cash

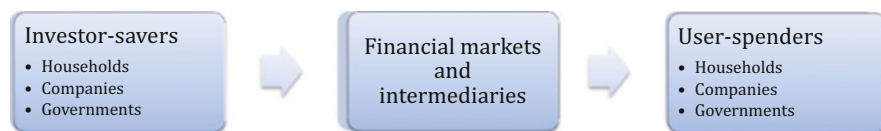


Fig. 4.2 Working of the financial system. *Source:* Adapted from De Haan et al. (2020)



Fig. 4.3 Visualising a stream of cash flows

flows. The *discount factor* then is the factor by which a future cash flow CF over n periods must be multiplied in order to obtain the present value:

$$\frac{1}{(1+r)^n} \quad (4.1)$$

Suppose the €100 today is equivalent to €103 in 1 year ($n = 1$). The discount rate is then 3%. And the €100 today is referred to as the present value of €103 in 1 year.

$$PV = \frac{CF_n}{(1+r)^n} \quad (4.2)$$

Another way to express this: the €103 in 1 year is the future value of €100 today. *Future value FV* is the value of a cash flow now, expressed in euros in the future.

$$FV = PV \cdot (1+r)^n \quad (4.3)$$

One can also calculate the present value of a stream of cash flows. This often includes negative cash flows as well, in which case one refers to the *net present value NPV*, i.e. net of cash outflows. Let's take the example of a 6-year bond (a debt security issued by governments or companies to investors). Figure 4.3 visualises the annual cash flows of the bond, from the perspective of the bondholder (the investor).

The negative signs are for cash outflows, in this case the payment of the bond's principal (of €1000) in 2022 to the company or country issuing the bond. The principal or face value of a bond is the amount the investor pays to the issuer of the bond. In the years 2023–2028, a €30 coupon (reflecting an annual interest rate payment of 3% of €1000) is received, and in 2028 the principal is paid back by the issuer. Let's suppose the discount rate of the bond is 3%, equal to the coupon. The NPV of the bond is then calculated as presented in Table 4.1. It starts by multiplying each annual cash flow with its associated discount factor. Remember that the discount factor is $1/(1+r)^n$. If 2022 is right now, then $n = 0$ and its discount factor is $1/(1+r)^n = 1/(1.03)^0 = 1$. For 2023, the discount factor is $1/(1+r)^n = 1/(1.03)^1 = 0.971$. Multiplying the annual cashflows with their discount factors gives the annual present values. The sum of those PVs is the NPV. The NPV of a cash flow stream from $n = 0$ to N then becomes:

$$NPV = \sum_{n=0}^N \frac{CF_n}{(1+r)^n} \quad (4.4)$$

Table 4.1 NPV of a stream of cash flows

Year	2022	2023	2024	2025	2026	2027	2028
Cash flow	−1000	30	30	30	30	30	1030
Discount factor	1	0.971	0.943	0.915	0.888	0.863	0.837
PV	−1000	29.1	28.3	27.5	26.7	25.9	862.6
NPV	0						

It might seem a coincidence that the NPV is exactly zero. However, in competitive markets, NPVs will often be zero or close to zero: competition between investors (arbitrage) will quickly result in the adjustment of prices of overpriced or underpriced securities (see Box 4.1 on arbitrage). In the above case, the NPV should be exactly zero, since the discount rate equals the coupon rate.

Box 4.1: Arbitrage and the Law of One Price

Arbitrage involves the buying and selling of the same or similar goods in different markets to benefit from price differences between these markets. Such arbitrage opportunities work if the law of one price does not hold. This ‘law’ says that the same (that is identical) products should sell at the same price. And if they don’t sell at the same prices, the arbitrage mechanism usually makes sure that such differences disappear quickly. For example, if an ounce of gold sells for \$2000 in country A and for \$1800 in country B, then a trader can benefit from the price difference, by buying gold in country B and selling it in country A. Their profit will be \$200 (\$2000−\$1800) minus the transaction costs that they incur in making the trade, such as transportation costs and taxes. If those transaction costs are less than \$200, they have an incentive to do the trade. And as many traders spot this opportunity, they will drive down the price differential to approximately the transaction costs involved.

Interest rates are often news items. A headline might say that central banks keep interest rates low; or that low interest rates hurt banks and push up housing prices. How does that work? Central banks indeed have a crucial role in setting interest rates: they provide short-term money to the market at a certain interest rate, which is called the *policy rate*. Over the past years, many central banks have in addition bought large amounts of bonds, driving up the prices of those bonds and lowering their effective annual interest rates at longer maturities (see further below for calculations). Box 4.2 provides an overview of the principal financial markets.

On June 21, 2021, the following data could be found on CNBC, a provider of financial market news and data, about the Brazilian 10-year government bond *yield* (that is the expected return on a bond when held till maturity), as Table 4.2 illustrates:

Based on the above information, we can derive the cash flow schedule as in the top row of Table 4.3 for the bond. If the bond was priced at par (i.e. at 1000) and the

yield went up. Prices and yields can change for various reasons. We will discuss this in Sect. 4.2.

Compounding

The above example showed how the present value of a stream of cash flows can be calculated, and how that present value may change in relation to the discount rate used. Another matter is the development of value over time as a return is earned on it.

Let’s suppose that you have €30,000 in a savings account and earn a 2% annual interest on it. What will be the value of your savings account in 50 years’ time? The answer depends on what you do with the intermediate returns. 2% on €30,000 is €600. So, if you cash in on your interest each year, then you will earn €600 each year and the value of your account stays at €30,000. The €600 is the simple interest: interest without the effect of compounding. However, if you keep the interest in your savings account, then your annual interest receipts will rise over time since the interest is calculated over a growing amount of capital. You then receive interest not just on the original amount of €30,000 but also on the interest previously earned. This is called compounded interest. The difference in value grows exponentially over time, as Table 4.7 illustrates.

With compounded interest, the value after 50 years is €80,748 instead of €30,000. Of course, a fairer comparison is to add the received interest ($50 \times €600 = €30,000$), in which case the value is €60,000. This is still €20,748 short, i.e. earned from interest on interest. People tend to underestimate this power of compounding. The effects are especially striking at higher returns and over longer periods of time. Figure 4.4 shows that interest on interest is insignificant at the beginning, but grows to 25% of total value (€20,748 out of €80,748).

The compounding effect is also visible when comparing rates of compounding. As seen in Fig. 4.4, 50 years of compounding at 2% results in a final value of €80,748 and a total return of 169%. But compounding at 4% gives a final value of €213,201 (2.6 times higher) and a total return of 611% (3.6 times higher). At 8% compounding, the final value is €1.4 million, over 17 times higher than at 2% (Table 4.8).

Table 4.7 Capital with and without compounding

Year	2% not compounded		2% compounded	
	Capital	Return	Capital	Return
1	30,000	600	30,600	600
2	30,000	600	31,212	612
3	30,000	600	31,836	624
4	30,000	600	32,473	637
5	30,000	600	33,122	649
...				
49	30,000	600	79,164	1552
50	30,000	600	80,748	1583

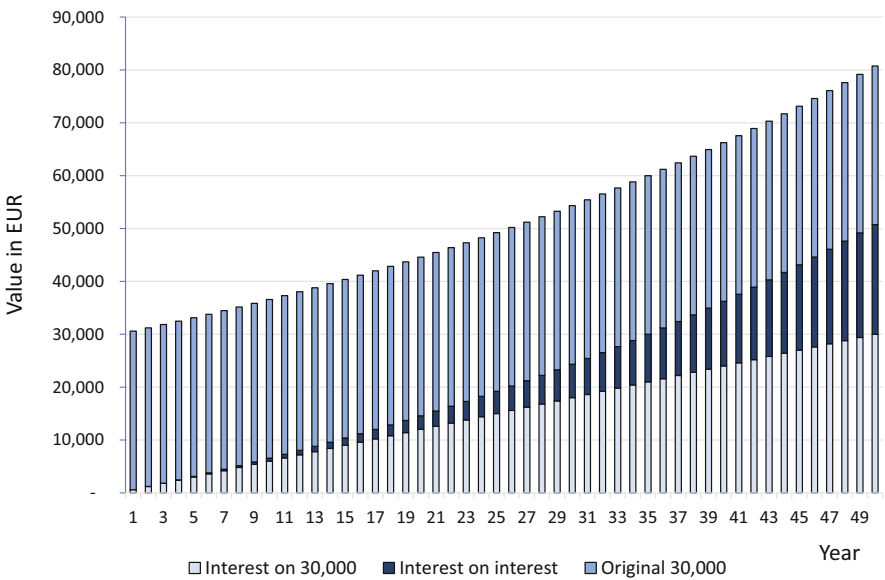


Fig. 4.4 Value composition with compounding returns

Table 4.8 Final values over different time periods and at different compounding rates

Annual return	Years				
	10	20	30	40	50
2%	36,570	44,578	54,341	66,241	80,748
4%	44,407	65,734	97,302	144,031	213,201
8%	64,768	139,829	301,880	651,736	1,407,048

Perpetuities

A *perpetuity* is a stream of regular and equal cash flows into infinity. For example, a 3% perpetual bond may pay a €30 coupon on an annual basis forever, as Fig. 4.5 illustrates.

The principal of €1000 is never repaid as such, but its value is earned back by means of the perpetual stream of €30 per year. The value of a perpetuity can be calculated as follows:

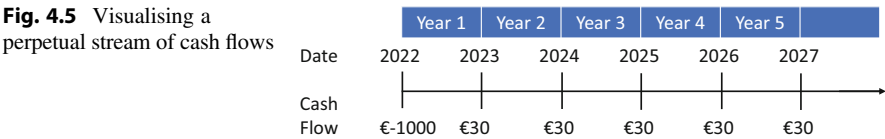


Fig. 4.5 Visualising a perpetual stream of cash flows

$$PV = \frac{CF_1}{(1+r)} + \frac{CF_2}{(1+r)^2} + \frac{CF_3}{(1+r)^3} + \dots = \sum_{n=1}^{\infty} \frac{CF_n}{(1+r)^n} \quad (4.5)$$

which can be reduced¹ to:

$$PV = \frac{CF}{r} \quad (4.6)$$

In the above example, the value of the perpetuity is $\text{€}30/0.03 = \text{€}1000$. This outcome satisfies the *law of one price*: the value of the perpetuity equals the cost to create it.

In practice, perpetual bonds do end at some point as their issuer ceases to exist. Nevertheless, some survive for centuries. For example, Yale owns a Dutch water bond from 1648 that still pays interest.²

Annuities

Like a perpetuity, an *annuity* is a stream of equal cash flows paid at regular intervals. Unlike a perpetuity, an annuity ends after a predetermined number of payments. So, an annuity is effectively a perpetuity with an end date ($n = N$) and its calculation reflects that: it is a perpetuity minus that same perpetuity with a later start date (and hence discounted). The value of an annuity can be calculated as follows:

$$PV = \frac{CF}{r} \cdot \left(1 - \frac{1}{(1+r)^N} \right) \quad (4.7)$$

Annuities are often used in mortgages to smooth out the interest payments that are higher at the beginning than later on (when the residual amount has fallen). The borrower pays then equal amounts (divided over interest payment and principal repayment) over the lifetime of the mortgage.

¹The derivation is as follows (Brealey et al., 2023). We start with Eq. (4.5): $PV = \frac{CF_1}{(1+r)} + \frac{CF_2}{(1+r)^2} + \frac{CF_3}{(1+r)^3} + \dots$. Now let $\frac{CF}{1+r} = a$ and $\frac{1}{1+r} = x$. Then we get (4.5a): $PV = a \cdot (1 + x + x^2 + \dots)$. Multiplying both sides by x , we get (4.5b): $PV \cdot x = a \cdot (x + x^2 + \dots)$. Subtracting (4.5b) from (4.5a), we get $PV \cdot (1 - x) = a$. Substituting back for a and x , we get $PV \cdot \left(1 - \frac{1}{1+r} \right) = \frac{CF}{1+r}$. Multiplying both sides by $(1+r)$ and then dividing by r gives Eq. (4.6): $PV = \frac{CF}{r}$.

²A living artifact from the Dutch Golden Age: Yale's 367-year-old water bond still pays interest | YaleNews

4.2 Determinants of Discount Rates on Financial Capital

So far, we have taken discount rates as a given. But what discount rate should investors use when discounting their expected cash flows? One could of course consider market interest rates—but there are many different interest rates, and there is no single market interest rate. Moreover, the appropriate discount rate depends on a number of issues, some of them specific to the investor or the investment. It therefore makes sense to use a discount rate based on the investor's opportunity cost of capital, also known as the cost of capital. This is the best available return on an investment that has risk and conditions similar to the cash flows to be discounted—either in the market or in other projects available to the company. Box 4.3 illustrates the opportunity cost of capital.

Box 4.3: Opportunity Cost of Capital

If a project is expected to return 11% to the company, while similar investments in the market deliver 9%, then the opportunity cost of capital is 9% and the project promises an excess return of 200 basis points. However, the dynamics can change if the company has several competing projects and a scarcity of capital. For example, if the company has a competing project that is expected to return 14%, then the opportunity cost of capital of the first project is 14% and the excess return of the 11% project is –300 basis points. In that case, the 11% project should not be done, and the 14% project should be chosen. Of course, both should be done if there is sufficient capital to do both projects.

For risk-free projects, a benchmark for (minimal) market risk can be chosen, such as the highest-rated government bonds. However, most corporate projects are not risk-free but carry additional risks that raise the discount rate. Investors typically determine those risk markups (or risk premiums) over the risk-free rate and the resulting discount rate by means of formulas and/or rules of thumb. For example, a private equity investor might use a standard 20% cost of equity for early-stage investments and then apply discounts (deductions) or premia (further markups) on this standard cost for specific industries, more or less mature technologies, quality of management, etc.

There are many determinants of discount rates, such as the demand and supply of capital; expected inflation; investment horizon; tax deductibility of interest rates; seniority; credit risk; illiquidity; sustainability; and behavioural aspects. Analytically, these determinants are typically split in the components that drive government bond yields (benchmark rate); and those that drive the premia on top of them, such as the corporate bond premium and the equity premium (Fig. 4.6). A brief overview is provided here. Chapter 8 on bonds and Chap. 9 on public equities provide a more in-depth treatment of the bond premium (also called the yield spread) and the equity premium.

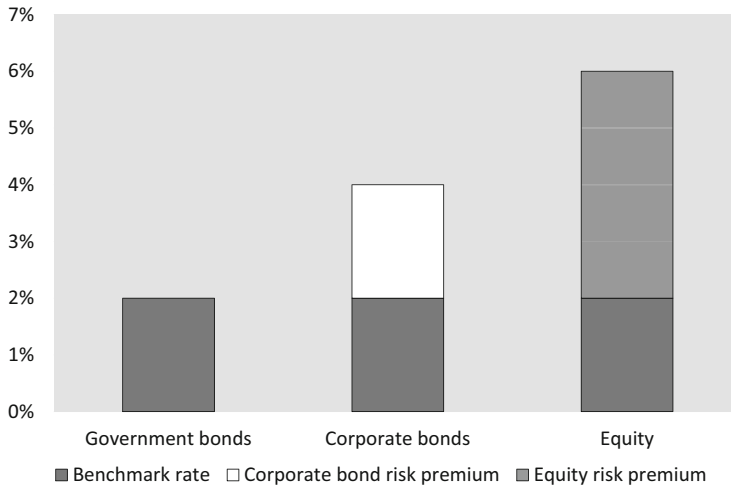


Fig. 4.6 Discount rates and risk premia (illustrative numbers)

Benchmark: Government Bonds

Market discount rates refer to the interest rates on government securities in specific markets. The highest quality government securities are considered ‘risk-free’. Market discount rates are the benchmarks against which discount rates are determined. If they move, then all discount rates move. Such moves are effectively shifts in the scarcity of capital. When supply of capital is tight and demand is high, market discount rates (the price of capital) will be high. Conversely, when supply is ample, such as in the recent period of quantitative easing, market discount rates will be low and sometimes even negative. The scarcity is partly set by the central bank’s supply of money. But the scarcity is also a function of the risk appetite in the market and the need for capital.

The investment horizon is another consideration. Short-term interest rates are strongly influenced by the monetary policy of central banks. Box 4.4 explains the different money market segments for funds up to 1 year. Yields of government bonds are influenced by expected short-term interest rates and the term premium. Risk-averse investors demand a *term premium* (or *risk premium*) for investments in long-term bonds to compensate them for the risk of losses due to (unexpected) interest rate hikes; those losses increase with a bond’s *duration*, which is the sensitivity of a bond’s price to changes in interest rates (see Chap. 8).

Box 4.4: Money Market Segments

Money markets are split into:

- Unsecured segment: money borrowed without collateral. This is subject to credit risk and will demand a credit risk premium.
- Secured segment: repurchase agreements (repos) whereby asset is sold against money, while the seller has right and obligation to repurchase the asset at specific price on future date. The underlying asset serves as collateral, eliminating credit risk.

The term premium leads to a positive *term spread*, i.e. the difference or spread between yields for bonds with longer maturity and yields for bonds with shorter maturity. The term spread can even be positive when markets expect increasing and decreasing interest rates to be equally likely.³ A positive term spread reflects what is often called a ‘normal’ yield curve. Figure 4.7 shows such an upward sloping yield

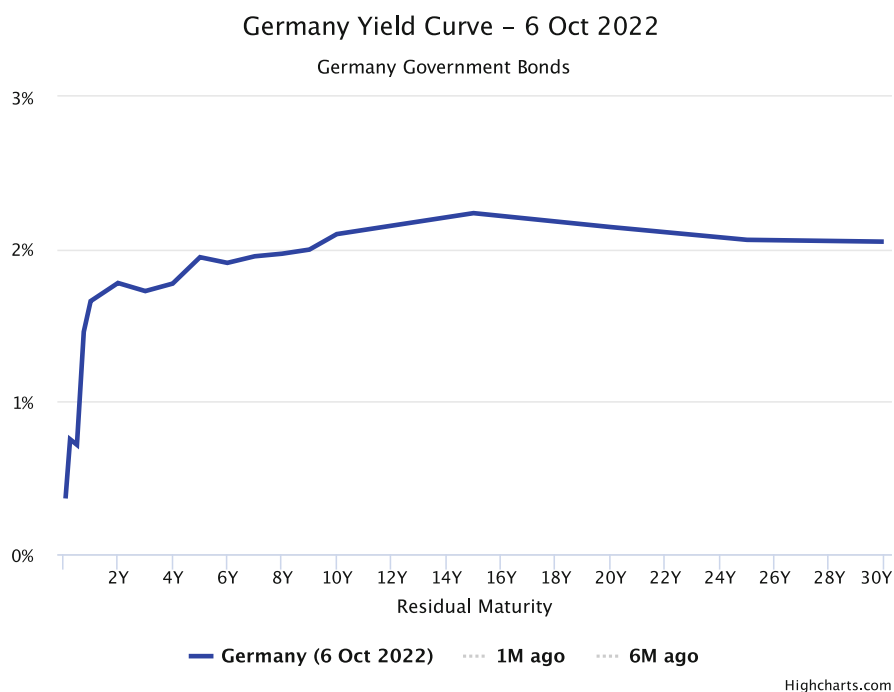


Fig. 4.7 German government bond yield curve. Source: World Government Bonds

³The term premium can also be negative providing a downward sloping yield curve (inverted yield curve). This depends on the investment horizon of investors: long-term investors, for example, may have a preference for long-term bonds resulting in a lower (or negative) term premium.

curve for German government bonds. A *yield curve* is a visualisation of the term structure, which is the relation between yields and maturities of otherwise similar bonds.

Apart from interest rate expectations and the term premium, *credit risk* and *liquidity* also influence government bond yields. Credit risk is the risk of loss because of the failure of a counterparty to perform according to a contractual arrangement, for instance due to a default by a borrower. The spread between the yield of a particular bond and the yield of a bond with similar characteristics but without credit risk is the *credit risk premium*. Rating agencies—like Moody's, Standard & Poor's (S&P), and Fitch—indicate issuers' credit risk by assigning them a *credit rating*. These are assessments of the risk of default. For example, at the time of writing (2022), Germany, the Netherlands, and the USA had AAA ratings and China and Japan had A+ ratings, whereas Brazil had a BB– rating and India was at BBB–. Differences in those credit ratings are driven by per capita income, GDP growth, inflation, external debt, level of economic development, and default history of the respective governments (Cantor & Pecker, 1996).

Liquidity is the ease with which an investor can sell or buy a bond immediately at a price close to the market price (see Box 8.2 in Chap. 8). Investors prefer more liquid securities, all else equal. The spread between the yield of a bond with high liquidity and a similar bond with less liquidity is referred to as the *liquidity premium*.

In Europe, Germany is the most creditworthy country. German government bonds form the deepest market (most liquid) and serve as benchmark for the euro-yield curve. In the USA, the US Treasury is the benchmark for the dollar-yield curve: see Fig. 4.8. As you can see, the US yield curve is above the German yield curve (Fig. 4.7) at the time of writing.

Corporate Bonds: Yield Spread

Corporate bonds tend to carry more serious default risk and lower liquidity than government bonds. These are the main drivers of the corporate yield spread, which is the difference between yields on corporate bonds and government bonds with the same maturity and rating. For example, yields of AAA-rated corporate bonds are generally higher than AAA-rated government bond yields. The corporate yield spread can be calculated per rating class and per maturity (Table 4.9).

Default risk is the risk that a bond will not make its promised payments. This is higher for corporate bonds since, unlike governments, they do not have the option of raising taxes to meet their payment obligations. They have to meet their obligations from their own cash flow. Defaults of corporate bonds are related to the business cycle and clustered in recession times. Bonds with higher operational and sustainability risk, and higher sensitivity to the business cycle, tend to have lower ratings and higher yields. Operational and sustainability risks can stem from social and environmental issues such as labour unrest or environmental costs that put a burden on financial performance (see Chap. 8 for more detail on default risk).

The lower liquidity of corporate bonds stems from lower trading frequencies and higher transaction costs, which are due to the smaller sizes of individual corporate bond markets (i.e. per security) and less competition among bond traders.

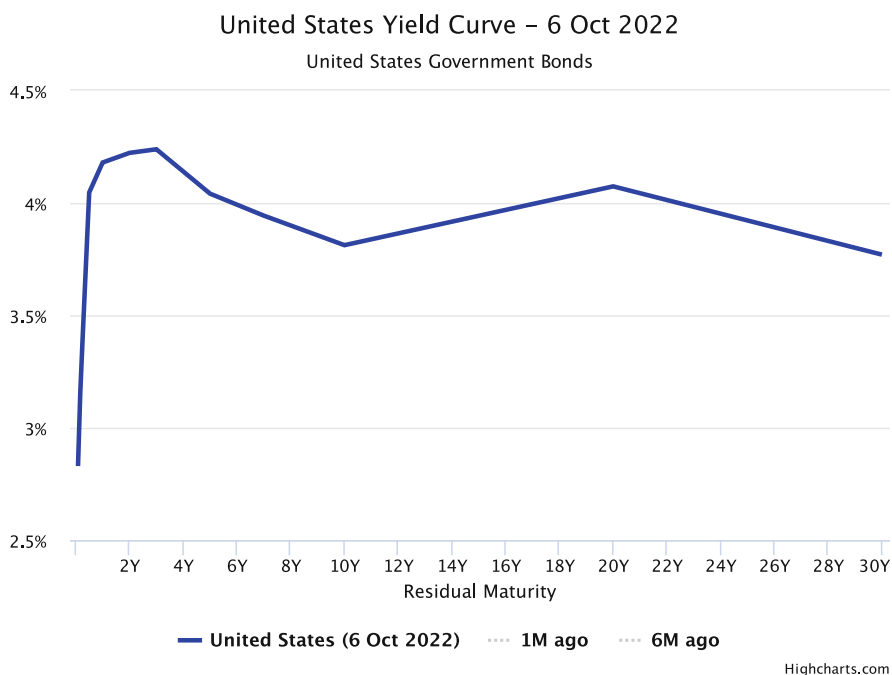


Fig. 4.8 US government bond yield curve. Source: World Government Bonds

Table 4.9 Illustrative corporate yield spread per maturity, November 2022

	1 year	5 years	10 years	20 years
AAA corporate bonds	4.39%	4.30%	4.39%	4.61%
AAA government bonds	4.07%	3.85%	3.65%	4.04%
AAA corporate yield spread	0.31%	0.45%	0.75%	0.57%

Source: Bloomberg

Equities: Market Risk Premium

Shareholders are the so-called residual claimants, in that they are paid out of the profits that remain after the other stakeholders and the providers of credit have been paid. As a result, equity typically carries a higher risk than corporate bonds. This higher risk is reflected in the equity risk premium, which is the expected excess return of equities over the risk-free rate. The equity risk premium is mostly related to market risk, but can also be related to social and environmental risk. The equity risk premium tends to be higher for smaller companies, more cyclical companies, and companies with weaker corporate governance. This is discussed more thoroughly in Chap. 12 on risk-return analysis.

Stock markets trade public equity, but equity can be private as well. In fact, most equity in most companies starts as private and remains private. That means there are high transaction costs involved in buying and selling it. As a result, private

companies typically have a liquidity discount over public companies (i.e., the equity value of private companies is lower than that of similar public companies).

Behavioural Explanations

The above sections described the determinants of discount rates from the perspective of the securities and markets under consideration. But discount rates are also affected by the ones using them, i.e. the investors. Investors may have behavioural biases that induce them to use other, lower or higher, discount rates than expected. For example, people have the tendency to use the same discount rate for all investment projects within a company, even if they differ significantly in their risk profile. This is called the ‘one discount rate fits all heuristic’.

4.3 Discounting Social and Environmental Capital

The previous section discussed the determinants of discount rates of financial capital, which come about in a setting of markets with supply and demand of financial funds. Figure 4.2 in Sect. 4.1 illustrates the flow of financial funds from investors to users through financial markets and intermediaries. A large supply of funds relative to demand lowers the price or discount rate of financial capital, *ceteris paribus*.

In a similar vein, social and environmental flows can be discounted as well. However, for social and environmental capital, the setting is different. The counterparty of companies’ social and environmental capital is the wider society, representing current and future generations. This raises two fundamental and ethical questions:

1. Should current and future generations be treated equally?
2. What is the appropriate discount rate for society (the social discount rate)?

Why Social Discount Rates Are Low

Many moral philosophers (e.g., Krznaric, 2020; Rawls, 1971) and economists (e.g. Ramsey, 1928; Stern, 2006) argue for an equal treatment of current and future generations. In economic terms, this means that the well-being of future generations gets the same weight as the well-being of the present generation (Dasgupta, 2021). So, when evaluating investment proposals for combatting climate change or preserving biodiversity, the well-being or ‘interest’ of future generations is taken fully into account alongside the interest of the present generation. This implies a zero-time preference between current and future generations. A positive time preference would advantage the present generation at the expense of future generations (facing global warming or biodiversity loss). Krznaric (2020, p.142) describes lucidly the practice of discounting the future as follows ‘Discounting is an iconic expression of the colonisation of the future, treating it as virtually empty of inhabitants’.

But there is more to social discounting than the time preference δ between current and future generations. Ramsey (1928) derives the social discount rate r^s for the appraisal of social projects:

$$r^s = \delta + \eta \cdot g \quad (4.8)$$

The growth rate g is driven by growth in consumption as well as by total factor productivity growth (i.e., innovations increase efficiency of production). The latter means that future generations have cheaper and more innovative solutions than those in the present. Given a diminishing marginal utility of consumption, the growth rate is multiplied by the elasticity of marginal utility of consumption η . The elasticity measures how utility changes with consumption. So, if future generations have better technology or if there is a greater elasticity, you should discount more. That is because the discount rate is set in such a way that consumers derive the same utility from current and future consumption. Higher future consumption thus increases the discount rate to equalise it with current consumption.

The social discount rate in Eq. (4.8) is a combination of time preference and (adjusted) consumption growth. The Ramsey rule measures the willingness to pay for a sure transfer of consumption through time. The risk premium for the social discount rate is introduced in Chap. 12.

Economists writing on global climate change (Cline, 1992; Nordhaus, 1994; Stern, 2006) have used Eq. (4.8) to identify optimum policies for correcting climate externalities (external impacts). Table 4.10 shows their different choice of parameters, assuming a growth rate of consumption of 1.3% (Stern, 2006). William Nordhaus is the outlier with a relatively large time preference of 3%. By contrast, William Cline and Nicholas Stern have a time preference close to zero, leading to discount rates of 1–2%. Dasgupta (2021) finds that the vast majority of economists find a social discount rate of 1–3% appropriate for long-run public projects. Low discount rates lead to high investment in combatting climate change, as the future benefits of restricting global warming are almost fully included because of limited discounting.

More broadly, low social discount rates imply that future environmental and social capital is almost as scarce as current environmental and social capital. Companies that add to these scarce capitals create integrated value and will be rewarded, while companies that draw these capitals down will face an increasing cost.

Table 4.10 Parameters for the social discount rate

Author	Social discount rate $r^s = \delta + \eta \cdot g$ with $g = 1.3\%$		
	Time preference δ	Elasticity η	Discount rate r^s
Cline (1992)	0%	1.5	1.95%
Nordhaus (1994)	3%	1	4.3%
Stern (2006)	0.1%	1	1.4%

Source: Dasgupta (2021)

4.4 Discounting Integrated Capital

Chapter 12 will more thoroughly address the issue of discounting on social and environmental capital (denoted by S capital or social value (SV) and E capital or environmental value (EV)) and add a risk parameter to the social discount rate. For now, let's assume 2% (the midpoint of Dasgupta's 1–3% range) for social and environmental capital for illustrative purposes. That makes calculations easier and means that discount rates on SV and EV are lower than on FV (financial value or capital). This in turn means that companies' discount rate on integrated capital rises with liabilities on SV and EV and falls with assets on SV and EV. Integrated capital is the combination of financial, social, and environmental capital. Let's see how this works.

If a company is investing in projects that contribute to scarce environmental or social capital, for example reducing carbon emissions or improving the health and safety of employees, it reduces its physical, transition or litigation risk (see Box 4.5). This lower risk reduces the integrated discount rate and thereby increases the value of the investment project. In Chap. 12, this is expressed as a low or negative beta on the environmental or social factor.⁴ By providing solutions for the SDGs, the company meets societal expectations. In contrast, a company that creates environmental or social liabilities, for example adopting high carbon technology or selling cigarettes, faces transition risk of future tightened policies or litigation risk of negligence of societal care. This raises the integrated discount rate.

Box 4.5: Breaking Down Environmental and Social Risks

Environmental risks, such as climate risk or risk of water shortages, and social risks, such as health and safety risks, can be broken down in several components:

- **Physical risks** are environmental events like floods or storms due to climate change or workplace injuries due to unsafe factories. Physical risks can affect companies directly through damage/loss of assets and injuries/deaths of employees, and indirectly through its effects on value chains and customers (see Chap. 1)
- **Transition risks** arise from changes in policy and new technologies. A carbon tax can affect the profitability of a company's business model (see Chap. 2) and
- **Litigation risk** is the risk that a company faces legal action. Examples are big litigation cases against tobacco companies on the health effects of smoking, and oil companies on not adhering to the Paris climate agreement of 2015 (see Chap. 12)

⁴Chapter 12 distinguishes between idiosyncratic risk (which can be diversified) and systemwide risk. Social and environmental risks have both systemwide dimensions and idiosyncratic dimensions; only the systemwide dimension is priced.

Table 4.11 Financial balance sheet of a standard company (only FV)

	Value	Discounted at		Value	Discounted at
F net operating assets	100	8.0%	F debt	20	4.0%
			F equity	80	9.0%
F capital	100	8.0%	F capital	100	8.0%

Table 4.12 Integrated balance sheet, company A (FV and EV)

	Value	Discounted at		Value	Discounted at
F net operating assets	100	8.0%	F debt	20	4.0%
E net assets	20	2.0%	F equity	80	9.0%
			E equity	20	2.0%
Integrated capital	120	7.0%	Integrated capital	120	7.0%

Table 4.13 Integrated balance sheet, company B (FV and EV)

	Value	Discounted at		Value	Discounted at
F net operating assets	100	8.0%	F debt	20	4.0%
E net assets	-20	2.0%	F equity	80	9.0%
			E equity	-20	2.0%
Integrated capital	80	9.5%	Integrated capital	80	9.5%

The starting point of our analysis is the financial balance sheet of a standard company, which only represents financial value. Let's assume the standard company has 100 in financial assets, financed by debt of 20 at 4% and equity of 80 at 9%. Table 4.11 shows the financial balance sheet of this standard company. The cost of capital of 8% is a weighted average of the cost of debt (4%) and the cost of equity (9%): $8\% = (20 \cdot 4\% + 80 \cdot 9\%)/100$ (see Chap. 13 for the weighted average cost of capital).

We now add environmental capital to construct an integrated balance sheet and leave out social capital for simplicity (Chap. 5 shows how environmental and social value can be calculated). Table 4.12 provides the balance sheet of company A which has positive net assets on EV (the company operates within planetary boundaries and helps other companies to reduce carbon emissions). The cost of integrated capital is now reduced to $7\% = (20 \cdot 4\% + 80 \cdot 9\% + 20 \cdot 2\%)/120$. The cost of integrated capital is lower than the cost of financial capital of 8%.

Instead of creating positive net assets, the standard company can also create negative net assets or liabilities. Table 4.13 shows the integrated balance sheet of company B with negative net assets on EV (not operating within planetary boundaries). The integrated capital is reduced to 80. The cost of integrated capital increases to $9.5\% = (20 \cdot 4\% + 80 \cdot 9\% - 20 \cdot 2\%)/80$. Hence, the integrated cost of capital of 9.5% is higher than the cost of financial capital of 8%.

We can do the same exercise for social capital, leaving out environmental capital for simplicity. An example of positive net assets on SV is investment in workplace

Table 4.14 Integrated balance sheet, company C

	Value	Discounted at		Value	Discounted at
F net operating assets	100	8.0%	F debt	20	4.0%
S net assets	10	2.0%	F equity	80	9.0%
			S equity	10	2.0%
Integrated capital	110	7.5%	Integrated capital	110	7.5%

Table 4.15 Integrated balance sheet, company D

	Value	Discounted at		Value	Discounted at
F net operating assets	100	8.0%	F debt	20	4.0%
S net assets	−10	2.0%	F equity	80	9.0%
			S equity	−10	2.0%
Integrated capital	90	8.7%	Integrated capital	90	8.7%

safety procedures. A chemical company can be a frontrunner in workplace safety. The company can use its safety technology for its own factories and at the same time provide safety consultancy services for other industries. Table 4.14 provides the integrated balance sheet of company C that has positive net assets on SV. The cost of integrated capital is reduced to $7.5\% = (20 \cdot 4\% + 80 \cdot 9\% + 10 \cdot 2\%) / 110$, which is lower than the cost of financial capital of 8%.

A company can also create social liabilities, for example selling cigarettes that cause health care problems. Table 4.15 depicts the integrated balance sheet of company D with negative net assets on SV (not operating within social boundaries). The cost of integrated capital increases to $8.7\% = (20 \cdot 4\% + 80 \cdot 9\% - 10 \cdot 2\%) / 90$, which is higher than the cost of financial capital of 8%.

Internalisation

The financial balance sheets for all the above companies look the same now, but will be different after internalisation of social and environmental externalities (e.g. through regulation) or the anticipation of internalisation (e.g. taking provisions for future litigation), as discussed in Chap. 2. The social discount rate is developed to analyse public investment projects. We apply it in this section to show scenarios in which companies want to (or must) meet societal expectations. The empirical prediction is that companies with large social and environmental liabilities will have a higher cost of integrated capital, *ceteris paribus*. This risk premium will rise when the risk of internalisation—that is the likelihood of (future) regulation or change in technology or consumer preferences—rises. By contrast, companies with social and environmental assets will enjoy a lower cost of integrated capital. Chapters 12 and 13 provide an in-depth analysis of the cost of integrated capital and Chap. 15 introduces integrated balance sheets. The cost of integrated capital can be used to discount projects on an integrated basis, including the financial, social, and environmental value dimensions.

As suggested in Chap. 2, *scenario analysis* is a useful tool to get insight into the possible internalisation of external impacts (externalities). In the scenario analysis, a

company can identify the most important societal trends for the industry in which it operates (including the likely internalisation of the main impacts) and assess its relative position (for example, degree of pollution or payment of living wages) within the industry.

4.5 Conclusions

When balancing capitals in investment decisions, one necessarily deals with the future. This raises the subject of this chapter: present values and discount rates. These are the basics of corporate finance. The first section of this chapter therefore discusses discount rates and the time value of money. Using the example of a Brazilian government bond, these concepts and the effects of changes in discount rates are shown in calculation examples. In addition, concepts such as compounding, perpetuities, and annuities are discussed.

Subsequently, the determinants of discount rates are discussed, starting from government bonds as a benchmark, and then adding the premia on corporate bonds and equity. This all applies to financial capital. Next, we introduce the social discount rate for social and environmental capital. The counterparty of companies' social and environmental capital is the wider society, representing current and future generations. Leading economists argue for an equal treatment of current and future generations, which implies a low social discount rate.

Finally, we show how the financial discount rate can be expanded to an integrated discount rate that can be applied to integrated value which also includes social and environmental value. It is shown that larger environmental and social liabilities raise the cost of integrated capital. Conversely, environmental and social assets lower the cost of integrated capital.

Key Concepts Used in This Chapter

Annuity is a stream of equal cash flows paid at regular intervals, which ends after a predetermined number of payments.

Arbitrage refers to the buying and selling of the same or similar goods in different markets to benefit from price differences between these markets (see also *law of one price*).

Bond market refers to the market segment for debt securities with a maturity of more than 1 year (bonds).

Compounding refers to the process whereby interest is credited to an existing principal amount as well as to interest already paid. Compounding is also referred to as interest on interest—the effect of which is to magnify returns to interest over time.

Credit rating refers to the assessment of the credit risk of prospective debtors by a rating agency, predicting their ability to pay back the debt.

Derivatives market refers to the place for trading derivatives, which are financial instruments whose value is derived from the value of the underlying financial instruments.

Discount factor is the factor by which a future cash flow must be multiplied in order to obtain the present value.

Discount rate refers to the interest rate used to determine the present value of future cash flows.

Equity market is the place where companies raise funds by issuing equity (that grants the investor a residual claim on the company's income) and investors trade equities.

Foreign exchange market is the place where traders buy and sell foreign currencies.

Future value refers to the value of a cash flow now, expressed in euros in the future.

Interest rate is the amount charged (interest) on top of the principal by a lender to a borrower for the use of financial funds.

Law of one price says that similar products should sell at the same price. And if they don't sell at similar prices, the arbitrage mechanism usually makes sure that such differences disappear quickly.

Liquidity is the ease with which an investor can sell or buy a bond immediately, at a price close to the market price

Money market is the market for short-term funds up to one year.

One discount rate fits all heuristic is the tendency to use the same discount rate for all investment projects within a company, even if they differ significantly in their risk profile.

Opportunity cost of capital refers to the best available return on an investment that has risk and conditions similar to the cash flows to be discounted—either in the market or in other projects available to the company.

Perpetuity is a stream of regular and equal cash flows into infinity.

Policy rate refers to the interest rate at which a central bank provides short-term money to the market.

Present value refers to present or current value of a discounted stream of future cash flows.

Social discount rate is the discount rate for social projects and can be used to discount social and environmental capital.

Term spread is the spread of yields for bonds with longer maturity over yields for bonds with shorter maturity.

Time value of money refers to people's preference for money today over money tomorrow.

Yield is the return on a bond.

Suggested Reading

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